DATA ITEM DESCRIPTION

Title: REPAIR WELDING APPROVAL RECORD FOR FORGINGS

Number: DI-FORG-81195A
AMSC Number: 10587
DTIC Applicable: No
Preparing Activity: MR
Applicable Forms: N/A

Approval Date: 20250721
Limitation: N/A
GIDEP Applicable: No
Project Number: FORG-2025-004

Use/relationship: The Repair Welding Approval Record for Forgings provides repair welding process and welder qualifications data. The Repair Welding Approval Record for Forgings is used to determine whether repair welding of forgings should be authorized.

a. This DID satisfies the requirements of paragraph 3.6.2 of MIL-DTL-46172.

b. This DID supersedes DI-FORG-81195.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The Repair Welding Approval Record for Forgings format shall be contractor selected. Unless a specific format is essential for compatibility, the initially used format arrangement by the contractor shall be used for all subsequent submissions.

3. Content. The Repair Welding Approval Record for Forgings content shall be in accordance with the elements of figure 1 and the Supplementary Data as follows:

3.1 Record number. The record shall include the contractor assigned (sequential and non-duplicative) record number with appropriate revision letter.

3.2 Forging Name. The record shall include the name of the forging.

3.3 Dwg. (Drawing) Number. The record shall cite the applicable Drawing Number.

3.4 Contract Number. The record shall include the applicable contract number.

3.5 Date. The record shall include the preparation date.

3.6 Welding Process. The record shall contain the welding process used.

3.7 Base Metals. The record shall include the Material Specification, the type or grade of the base metal(s), the Thickness Range, and the Diameter.

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3.8 Filler material. When applicable: the record shall include the weld metal type, the size of the electrode or filler, the MIL (military) Specification, the AWS (American Welding Society) classification, the electrode flux class, and the consumable insert.

3.9 Gas. When applicable: the record shall list the type of gas or gases, composition of gas mixture, orifice or gas cup size, shielding gas, flow rate, gas backing, and trailing shielding gas composition.

3.10 Position. The record shall identify the position of the groove, weld progression (uphill, downhill), method of backgouging.

3.11 Preheat. The record shall include the Preheat Temp (temperature), Interpass Temp (temperature), and Preheat Maintenance.

3.12 Electrical characteristics. The record shall identify the Current (AC or DC), Polarity (Straight or Reversed), Amps (amperage), Volts (voltage), and if pulsed welding techniques (GMAW-P) were used.

3.13 Technique. When applicable: the record shall identify the technique Travel Speed, String or Weave Bead, Oscillation, Multipass or Single Pass (how many passes per side), Single or Multiple Electrodes, Initial and Interpass Cleaning.

3.14 Post Heat. When applicable: the record shall include the post heat Temperature and Time.

3.15 Heat treatment after welding. The record shall specify the heat treatment after welding temperature (degrees Fahrenheit), time (minutes) and Quench Media for both austenitizing and tempering.

3.16 Manufacturer. The record must contain the name of the manufacturer (contractor).

3.17 Signature. The completed record must include the manufacturer's (contractor's) representative's name and signature.

4. Supplementary data. When the DD Form 1423 is not used in the forging contract or purchase order thus necessitating the use of this DID, the following supplementary data shall be included:

4.1 Weld size and location. A dimensioned sketch illustrating the size and location of the weld(s)

4.2 Weld repair procedure. Complete information on the weld repair procedure.

4.3 Welder qualifications. Complete information on the welder's qualifications.

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Record Number: _____

Forging Name: _____
Contract Number: _____
Date: _____

Welding Process: _____

Types (Manual, Automatic, Semi-Auto): _____

BASE METALS

Material Spec: _____
Type or Grade: _____
Thickness Range: _____
Diameter: _____

PREHEAT

Preheat Temp: _____
Interpass Temp: _____
Preheat Maintenance: _____
Remarks/Other: _____

FILLER MATERIAL

Weld Metal Type: _____
Size of Electrode or Filler: _____
MIL Specification: _____
AWS Classification: _____
Electrode Flux Class: _____
Consumable Insert: _____
Remarks/Other: _____

ELECTRICAL CHARACTERISTICS

Current (AC or DC): _____
Polarity (Straight or Reverse): _____
Amps: _____ Volts: _____
GMAW-P (yes/no): _____
Remarks/Other: _____
_____

GAS

Type of Gas or Gases: _____
Composition of Gas Mixture: _____
Orifice or Gas Cup Size: _____
Shielding Gas: _____
Flow Rate: _____
Gas Backing: _____
Trailing Shielding Gas: _____
Composition: _____
Remarks/Other: _____
_____

TECHNIQUE

Travel Speed: _____
String or Weave Bead: _____
Oscillation: _____
Multipass or Single Pass (per side): _____
Single or Multiple Electrodes: _____
Initial and Interpass Cleaning
(Brushing, Grinding, etc.): _____

POST HEAT

Temperature (°F): _____ Time: _____
Remarks/Other: _____

POSITION

Position of Groove: _____
Weld Progression
(Uphill, Downhill): _____
Remarks/Other: _____
Methods of Backgouging: _____
Remarks/Other: _____
_____

HEAT TREATMENT AFTER WELDING

Austenitize (°F): _____ Minutes: _____
Quench Media: _____
Tempering (°F): _____ Minutes: _____
Quench Media: _____

Manufacturer: _____

Signature: _____

Figure 1. Weld Repair Data and Weld Equipment Qualification Record.

End of DI-FORG-81195A.

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